

EAU GUIDELINES ON NON-MUSCLE-INVASIVE (TaT1, CIS) BLADDER CANCER

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Epidemiology

Bladder cancer (BC) is the 7th most commonly diagnosed cancer in the male population worldwide, while it drops to the 10th position when both genders are considered. The worldwide age-standardised incidence rate (per 100,000 person/years) is 9.5 in men and 2.4 in women.

Staging and classification systems

The 2017 TNM (Tumour, Node, Metastasis Classification) is used for staging (Table 1). For grading, both the **older 1973** (which distinguish between grade 1 [G1], grade 2 [G2] and grade 3 [G3] categories) and the newer **2004/2022 WHO** grading classifications (that categorises BC into papillary urothelial neoplasm of low malignant potential (PUNLMP), non-invasive papillary carcinoma low grade (LG) and high grade (HG)) are used. While the 2 grading systems are prognostic for progression, **a 3-tier hybrid** (LG/G1-G2, HG/G2 & HG/G3) or a **4-tier combination** (LG/G1, LG/G2, HG/G2 and HG/G3) both **proved to be superior to either classification system alone.**

Table 1: TNM Classification 2017

T - Primary Tumour	
TX	Primary tumour cannot be assessed
T0	No evidence of primary tumour
Ta	Non-invasive papillary carcinoma
Tis	Carcinoma <i>in situ</i> : 'flat tumour'
T1	Tumour invades subepithelial connective tissue
T2	Tumour invades muscle
T2a	Tumour invades superficial muscle (inner half)
T2b	Tumour invades deep muscle (outer half)
T3	Tumour invades perivesical tissue
T3a	Microscopically
T3b	Macroscopically (extravesical mass)
T4	Tumour invades any of the following: prostate stroma, seminal vesicles, uterus, vagina, pelvic wall, abdominal wall
T4a	Tumour invades prostate stroma, seminal vesicles, uterus or vagina
T4b	Tumour invades pelvic wall or abdominal wall
N – Regional Lymph Nodes	
NX	Regional lymph nodes cannot be assessed
N0	No regional lymph node metastasis
N1	Metastasis in a single lymph node in the true pelvis (hypogastric, obturator, external iliac, or presacral)
N2	Metastasis in multiple regional lymph nodes in the true pelvis (hypogastric, obturator, external iliac, or presacral)
N3	Metastasis in common iliac lymph node(s)

M - Distant Metastasis

M0	No distant metastasis
M1a	Non-regional lymph nodes
M1b	Other distant metastases

Carcinoma *in situ*

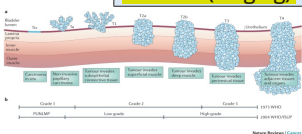
Carcinoma in situ (CIS) is a flat, high-grade, non-invasive urothelial carcinoma, and classified into the following clinical types:

- Primary: isolated CIS with no previous or concurrent papillary tumours and no previous CIS;
- Secondary: CIS detected during follow-up of patients with a previous tumour that was not CIS;
- Concurrent: CIS in the presence of any other urothelial tumour in the bladder.

Subtypes of urothelial carcinoma and lymphovascular invasion

Most subtypes of urothelial carcinoma (**micropapillary, plasmacytoid, sarcomatoid**) have a worse prognosis than pure high-grade (HG) urothelial carcinoma. The presence of lymphovascular invasion (LVI) in transurethral resection of the bladder (TURB) specimens is associated with worse prognosis.

Recommendations for bladder cancer classification	Strength rating
Use the 2017 TNM (Tumour, Node, Metastasis Classification) system for classification of the depth of tumour invasion (staging).	Strong



Provide T1 sub-stage if the lamina propria is adequately sampled using either micrometric (T1e and T1m) or histo-anatomic (T1a and T1b) principles.	Weak
Use both the 1973 and 2004/2022 World Health Organization (WHO) grading classification systems, or a hybrid system.	Weak
Do not use the term ' superficial' bladder cancer .	Strong

Diagnosis

A comprehensive patient history is mandatory. **Haematuria is the most common finding.** Carcinoma in situ can be suspected in patients with irritative voiding symptoms.

Recommendations for the primary assessment of non-muscle invasive bladder cancer	Strength rating
Take a patient history, focusing on urinary tract symptoms and haematuria.	Strong
Use <u>renal and bladder ultrasound and/or computed tomography-intravenous urography (CT-IVU)</u> during the initial work-up in patients with haematuria.	Strong
Once a bladder tumour has been detected, perform a CT urography in selected cases (e.g., tumours located in the trigone, multiple- or high-risk tumours).	Strong
Perform cystoscopy in patients with symptoms suggestive of bladder cancer or during surveillance. It cannot be replaced by cytology or by any other non-invasive test.	Strong

In men, use a flexible cystoscope, if available and apply irrigation 'bag squeeze' to decrease procedural pain when passing the proximal urethra.	Strong
Describe all macroscopic features of the tumour (site, size, number, and appearance) and mucosal abnormalities during cystoscopy. Use a bladder diagram.	Strong
Use voided urine cytology as an adjunct to cystoscopy to detect high-grade tumour.	Strong
Perform cytology on at least 25 mL fresh urine or urine with adequate fixation. First morning urine is not suitable because of the frequent presence of cytolysis.	Strong
Use the Paris System 2 nd Edn. for cytology reporting.	Strong

Papillary (TaT1) tumours

The diagnosis of papillary BC ultimately depends on cystoscopic examination of the bladder and histological evaluation of the tissue resected during TURB. Transurethral resection of the bladder is a crucial procedure in the diagnosis and treatment of TaT1 tumours and should be performed systematically in individual steps (see recommendations below). A complete resection, performed by either fractioned or *en-bloc* technique, is essential to achieve a good prognosis.

The technique selected depends on the size of the lesion, its location and experience of the surgeon. In selected cases, due to the risk of tumour persistence and understaging after initial TURB, a second resection (2nd TURB) is recommended.

Carcinoma *in situ*

Carcinoma *in situ* is diagnosed by a combination of cystoscopy, urine cytology, and histological evaluation of bladder biopsies taken from suspicious areas or as mapping biopsies from normal looking mucosa (for details, please consult the extended guidelines). Carcinoma *in situ* cannot be eradicated by TURB and further treatment is mandatory.

Recommendations for transurethral resection of the bladder (TURB), biopsies and pathology report	Strength rating
In patients suspected of having bladder cancer, perform a transurethral resection of the bladder tumour (TURB) followed by pathology investigation of the obtained specimen (s) as a diagnostic procedure and initial treatment step.	Strong

Perform TURB systematically in individual steps: • bimanual palpation under anaesthesia before starting the procedure and at the end; • insertion of the resectoscope, under visual control with inspection of the whole urethra; • inspection of the whole urothelial lining of the bladder; • biopsy from the prostatic urethra (if indicated); • cold-cup bladder biopsies (if indicated); • resection of the tumour; • recording of findings in the surgery report/record; including visual impression of grade/stage; • precise description of the specimen(s) for pathology evaluation.	Strong
Performance of individual steps	
Perform <i>en-bloc</i> resection or resection in fractions (exophytic part of the tumour, the underlying bladder wall and the edges of the resection area).	Strong
Avoid cauterisation as much as possible during TURB to avoid tissue deterioration.	Strong
Take biopsies from abnormal-looking urothelium.	Strong
Take multiple biopsies (mapping biopsies from the trigone, bladder dome, right, left, anterior and posterior bladder wall) or perform fluorescence-guided (PDD) biopsies, in case of normal looking urothelium and positive urine cytology.	Strong

Take a sample of the prostatic urethra if there is positive urine cytology without evidence of tumour in the bladder, or if abnormalities of the prostatic urethra are visible.	Strong
Take a sample biopsy of the prostatic urethra in cases of bladder neck tumour, suspicion of bladder carcinoma <i>in situ</i> (CIS) and/or T1 disease. If a sample was not taken during the initial procedure, it should be performed at the time of second resection, if the latter is needed.	Weak
Use methods to improve tumour visualisation (fluorescence cystoscopy, narrow-band imaging) during TURB, if available.	Weak
Refer the specimens from different biopsies and resection fractions to the pathologist in separately labelled containers. Submit the tumour base separately especially in large and multifocal tumours or when <i>en-bloc</i> resection is not feasible.	Weak
The TURB record must describe tumour location, appearance, size and multifocality, all steps of the procedure, extent, macroscopic completeness of resection as well as any complications.	Strong
In patients with positive cytology, but negative cystoscopy, exclude an upper tract urothelial carcinoma, CIS in the bladder (by mapping biopsies or PDD-guided biopsies) and tumour in the prostatic urethra (by prostatic urethra biopsy).	Strong

Perform a second TURB in the following situations: • after incomplete initial TURB, or in case of doubt about completeness of a TURB; • if there is no detrusor muscle in the specimen after initial resection, with the exception of Ta LG/G1 tumours and primary CIS; • in T1 tumours.	Strong
If indicated, perform a second TURB within two–six weeks after the initial resection. This second TURB should include resection of the primary tumour site.	Weak
Record the pathology results of the second TURB as it reflects the quality of the initial resection.	Weak
Inform the pathologist of prior treatments (intravesical therapy, radiotherapy, etc.).	Strong
The pathological report should specify tumour location, tumour grade and stage, lymphovascular invasion, subtypes of urothelial carcinoma, presence of CIS and detrusor muscle.	Strong

Predicting disease recurrence and progression and defining risk groups

After TURB, patients should be stratified, according to prognostic factors, into risk groups which will facilitate treatment recommendations (see Table 2). For individual prediction of the risk of tumour progression at different intervals after TURB, the use of the 2021 EAU NMIBC Risk Calculator (www.nmibc.net) is strongly recommended.

For bacillus Calmette-Guerin (BCG)-treated patients, separate scoring models and risk groups have been created by the CUETO and the EORTC, respectively. For prediction of tumour recurrence at 1 and 5 years in individual patients, the 2006 EORTC scoring model and calculator may be used (<https://www.omnicalculator.com/health/eortc-bladder-cancer>).

Recommendations for stratification of non-muscle invasive bladder cancer	Strength rating
Stratify patients into four risk groups according to Table 3. A patient's risk group can be determined by using the 2021 EAU risk group calculator available at www.nmibc.net .	Strong
For information about the risk of disease progression in a patient with primary TaT1 tumours not treated with bacillus Calmette-Guerin (BCG), use the data from Table 3.	Strong
Use the 2006 EORTC scoring model to predict the risk of tumour recurrence in individual patients not treated with BCG.	Strong
Use the 2016 EORTC scoring model or the CUETO risk scoring model to predict the risk of tumour recurrence in individual patients treated with BCG intravesical immunotherapy (the 2016 EORTC model is calculated for one to three year of maintenance, the CUETO model for five to six months of BCG).	Strong

Table 2: Clinical composition of the new EAU NMIBC prognostic factor risk groups based on the WHO 2004/2022 or the WHO 1973 grading classification systems

- Only one of the two classification systems (WHO 1973 or WHO 2004/2022) is required to use this table.
- If both classification systems are available in an individual patient, the Panel recommends using the risk group calculation based on the WHO 1973 as it has better prognostic value.
- The category of LG tumours (WHO 2004/2022) also includes patients with tumours classified as PUNLMP.
- Additional clinical risk factors are:
 - o age > 70;
 - o multiple papillary tumours;
 - o tumour diameter > 3 cm.

Risk Group	Description
Low Risk	• A primary, single, TaT1 LG/G1 tumour < 3 cm in diameter without CIS in a patient ≤ 70 years • A primary Ta LG/G1 tumour without CIS with at most ONE of the additional clinical risk factors
Intermediate Risk	Patients without CIS who are not included in either the low-, high-, or very high-risk groups

High Risk	• All T1 HG/G3 without CIS, EXCEPT those included in the very high-risk group • All CIS patients, EXCEPT those included in the very high-risk group Stage, grade with additional clinical risk factors: • Ta LG/G2 or T1 G1, no CIS with all 3 risk factors • Ta HG/G3 or T1 LG, no CIS with at least 2 risk factors • T1 G2 no CIS with at least 1 risk factor
Very High Risk	Stage, grade with additional clinical risk factors: • Ta HG/G3 and CIS with all 3 risk factors • T1 G2 and CIS with at least 2 risk factors • T1 HG/G3 and CIS with at least 1 risk factor • T1 HG/G3 no CIS with all 3 risk factors

The scoring model is based on individual patient data, but does not consider patients with primary CIS (high-risk) or with recurrent tumours, as well as some pathologic parameters like subtypes of urothelial carcinoma (*micropapillary, plasmacytoid, sarcomatoid, small-cell, neuroendocrine*) and LVI.

Nevertheless:

- Based on data from the literature, all patients with CIS in the prostatic urethra, with subtypes of urothelial carcinoma or with LVI should be included in the very high-risk group.
- Patients with recurrent tumours should be included in the intermediate-, high- or very high-risk groups according to their other prognostic factors.

Table 3: Probabilities of disease progression in 1, 5 and 10 year(s) for the new EAU NMIBC risk groups

Risk group	Probability of Progression and 95% Confidence Interval (CI)		
	1 Year	5 Years	10 Years
New Risk Groups with WHO 2004/2016			
Low	0.06% (CI: 0.01%–0.43%)	0.93% (CI: 0.49%–1.7%)	3.7% (CI: 2.3%–5.9%)
Intermediate	1.0% (CI: 0.50%–2.0%)	4.9% (CI: 3.4%–7.0%)	8.5% (CI: 5.6%–13%)
High	3.5% (CI: 2.4%–5.2%)	9.6% (CI: 7.4%–12%)	14% (CI: 11%–18%)
Very High	16% (CI: 10%–26%)	40% (CI: 29%–54%)	53% (CI: 36%–73%)
New Risk Groups with WHO 1973			
Low	0.12% (CI: 0.02%–0.82%)	0.57% (CI: 0.21%–1.5%)	3.0% (CI: 1.5%–6.3%)
Intermediate	0.65% (CI: 0.36%–1.2%)	3.6% (CI: 2.7%–4.9%)	7.4% (CI: 5.5%–10%)
High	3.8% (CI: 2.6%–5.7%)	11% (CI: 8.1%–14%)	14% (CI: 10%–19%)
Very High	20% (CI: 12%–32%)	44% (CI: 30%–61%)	59% (CI: 39%–79%)

WHO = World Health Organization.

Disease management

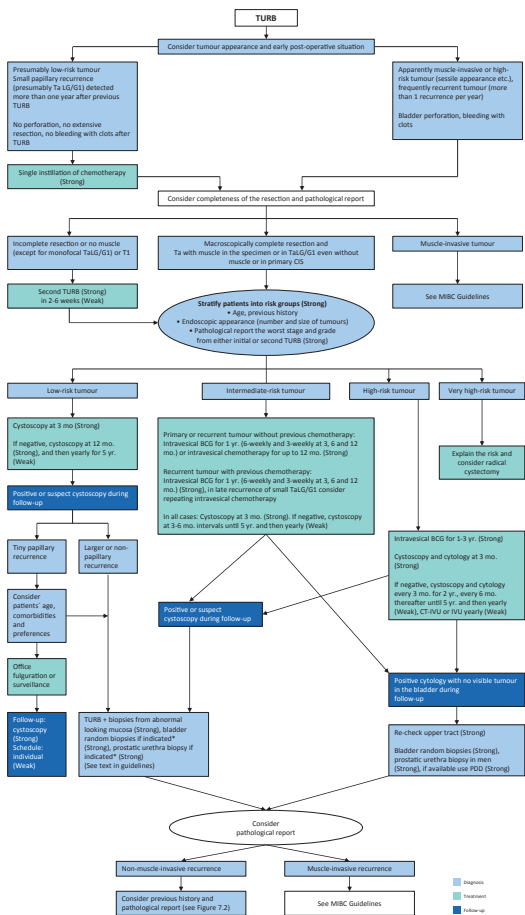
Adjuvant treatment

Since there is considerable risk for recurrence and/or progression of tumours after TURB, adjuvant intravesical therapy is recommended for all stages (TaT1, and CIS).

- **Immediate single post-operative instillation of chemotherapy** immediate single post-operative instillation of chemotherapy after TURB can reduce the recurrence rate in patients with low-risk and selected intermediate-risk tumours. The difference in efficacy between individual drugs (mitomycin C, epirubicin, or doxorubicin) has not been confirmed.
- **Further chemotherapy** instillations can improve recurrence-free survival in intermediate-risk tumours, but do not prevent progression. These instillations are associated with minor side effects.
- **Intravesical immunotherapy with BCG** (induction and maintenance) is superior to intravesical chemotherapy in reducing recurrences and in preventing or delaying progression to MIBC.

The individual choice of further intravesical adjuvant therapy depends on the patient's risk (Table 2). In patients at very high risk of progression, immediate radical cystectomy (RC) should be considered.

Figure 1: Treatment strategy in primary or recurrent tumour(s) without previous BCG*



* For details and explanations see the text of the extended NMIBC guidelines available at: <https://uroweb.org/guidelines/non-muscle-invasive-bladder-cancer>.

BCG = bacillus Calmette-Guérin; CIS = carcinoma in situ; CT = computed tomography; IVU = intravenous urography; MIBC = muscle-invasive bladder cancer; PDD = photodynamic diagnosis; TURB = transurethral resection of the bladder.

Bacillus Calmette-Guérin failure

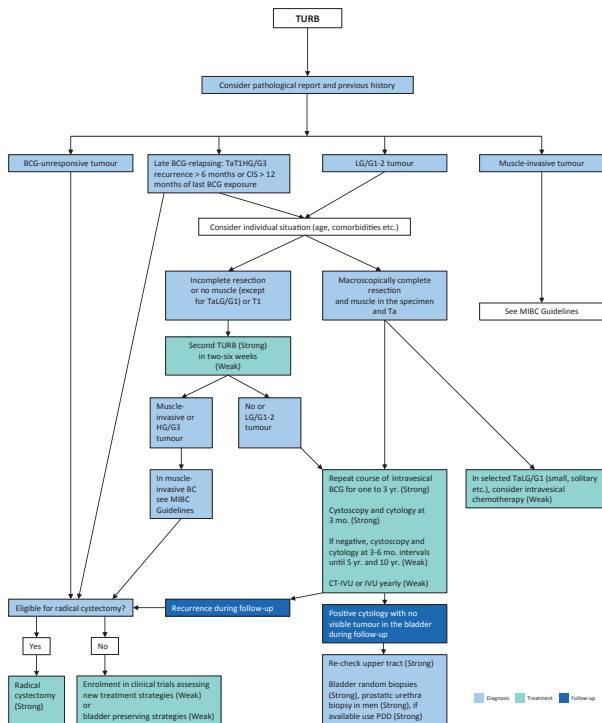
Several categories of BCG failures, broadly defined as any HG recurrence following BCG therapy, have been proposed.

Whenever a MIBC is detected during follow-up.
BCG-refractory tumour
1. If T1 HG/G3 tumour is present at 3 months (LE: 3).
2. If Ta HG/G3 tumour is present after 3 months and/or at 6 months, after either re-induction or first course of maintenance (LE: 4).
3. If CIS (without concomitant papillary tumour) is present at 3 months and persists at 6 months after either re-induction or first course of maintenance. If patients with CIS present at 3 months, an additional BCG course can achieve a complete response in > 50% of cases (LE: 1b).
4. If HG tumour appears during BCG maintenance therapy*.
BCG-relapsing tumour
Recurrence of HG/G3 tumour after completion of BCG maintenance, despite an initial response (LE: 3).
BCG unresponsive tumour
BCG unresponsive tumours include all BCG refractory tumours and those who develop T1/Ta HG recurrence within 6 months of completion of adequate BCG exposure** or develop CIS within 12 months of completion of adequate BCG exposure (LE: 4).

BCG-exposed tumour
If Ta HG/G3 or CIS is present at 3 months evaluation after induction BCG only.
Delayed relapse after adequate or inadequate BCG.
BCG intolerance
Severe side effects that prevent further BCG instillation before completing treatment.

- * *Patients with LG recurrence during or after BCG treatment are not considered to be a BCG failure.*
- ** *Adequate BCG is defined as the completion of at least 5 of 6 doses of an initial induction course plus at least 2 out of 6 doses of a second induction course or 2 out of 3 doses of maintenance therapy.*

Figure 2: Treatment strategy in recurrence during or after intravesical BCG



BCG = bacillus Calmette-Guérin; CIS = carcinoma in situ; HG = high-grade; IVU = intravenous urography; LG = low-grade; PDD = photodynamic diagnosis; TURB = transurethral resection of the bladder

General recommendations for adjuvant therapy in TaT1 tumours and for therapy of CIS	Strength rating
Counsel smokers to stop smoking.	Strong
The type of further therapy after transurethral resection of the bladder (TURB) should be based on the risk groups shown in Section 6.3 and Table 6.1. For determination of a patient's risk group use the 2021 EAU risk group calculator available at www.nmibc.net .	Strong
In patients with tumours presumed to be at low risk and in those with small papillary recurrences (presumably Ta LG/ G1) detected more than one year after previous TURB, offer one immediate single chemotherapy instillation.	Strong
Offer post-operative saline or water continuous irrigation of the bladder to patients who cannot receive a single instillation of chemotherapy.	Strong
Patients with small recurrent low-grade Ta tumours can be effectively and safely offered office fulguration.	Strong
Only offer active surveillance to selected patients with presumed low-risk tumours not amendable to endoscopic ablation.	Weak

In patients with intermediate-risk tumours (with or without immediate instillation), offer instillations of chemotherapy (the optimal schedule is not known) or one-year full-dose Bacillus Calmette-Guérin (BCG) treatment (induction plus 3-weekly instillations at 3, 6 and 12 months). Chemotherapy is a reasonable first option in the majority of cases; however, the final choice should be made in a shared decision-making process with the patient, reflecting his/her risk of recurrence and progression, as well as the efficacy and side effects of each treatment modality.	Strong
Administer a full-dose intravesical bacillus Calmette-Guérin (BCG) for one to three years in patients with high-risk tumours (a complete BCG schedule comprises an induction phase of six-weekly instillations, followed by a maintenance phase of three-weekly instillations at three, six, twelve, eighteen, 24, 30 and 36 months, respectively). The additional beneficial effect of the second and third years of maintenance should be weighed against its added costs, side effects and access to BCG. Immediate radical cystectomy (RC) may also be discussed with the patient.	Strong
Discuss immediate RC in patients with very high-risk tumours. Intravesical full-dose BCG instillations for one to three years remains an option for selected patients, particularly those who decline or are unfit for RC.	Strong

Offer transurethral resection of the prostate, followed by intravesical instillation of BCG to patients with CIS in the epithelial lining of the prostatic urethra if a bladder sparing strategy is considered.	Weak
Cautiously offer quinolones to treat BCG-related side effects*.	Weak
The definition of 'BCG unresponsive' should be respected as it most precisely defines the patients who are unlikely to respond to further BCG instillations.	Strong
Offer a RC to patients with BCG unresponsive tumours.	Strong
Offer patients with BCG unresponsive tumours, who are not candidates for RC due to comorbidities, preservation strategies (intravesical chemotherapy, chemotherapy and microwave-induced hyperthermia, electromotive administration of chemotherapy, intravesical- or systemic immunotherapy; preferably within clinical trials).	Weak
Discuss high-risk and very high-risk patients within a multidisciplinary board, when possible.	Weak
Recommendations – technical aspects for treatment	
<i>Intravesical chemotherapy</i>	
If given, administer a single immediate instillation of chemotherapy within 24 hours after TURB.	Weak

Omit a single immediate instillation of chemotherapy in any case of overt or suspected bladder perforation or bleeding requiring bladder irrigation.	Strong
The optimal schedule and duration of further intravesical chemotherapy instillation is not defined; however, it should not exceed one year.	Weak
If intravesical chemotherapy is given, use the drug at its optimal pH and maintain the concentration of the drug by reducing fluid intake before and during instillation.	Strong
The length of individual instillation should be a minimum of one, and up to two hours.	Weak
BCG intravesical immunotherapy	
Absolute contraindications of BCG intravesical instillation are: • during the first two weeks after TURB; • in patients with visible haematuria; • after traumatic catheterisation; • in patients with symptomatic urinary tract infection.	Strong

* The side-effect profile of quinolones and fluoroquinolones resulted in the adoption of European regulation restricting their use.

Recommendations for the treatment of TaT1 tumours and carcinoma <i>in situ</i> according to risk stratification	Strength rating
EAU risk group: Low	
Offer one immediate instillation of intravesical chemotherapy after transurethral resection of the bladder (TURB).	Strong

EAU Risk Group: Intermediate

In general, chemotherapy (the optimal schedule is unknown) is a reasonable first-line option in the majority of patients. One-year full-dose BCG treatment (induction plus three-weekly instillations at three, six and twelve months), is an alternative option. The final choice should reflect the individual patient's risk of recurrence and progression as well as the efficacy and side effects of each treatment modality. Offer one immediate chemotherapy instillation to patients with small papillary recurrences detected more than one year after previous TURB.

Strong

EAU risk group: High

Offer intravesical full-dose BCG instillations for one to three years but discuss immediate radical cystectomy (RC).

Strong

EAU risk group: Very High

Offer RC or intravesical full-dose BCG instillations for one to three years, particularly to those who decline or are unfit for RC.

Strong

Table 4: Treatment options for the various categories of BCG failure

Category	Treatment options
BCG-unresponsive	1. Radical cystectomy (RC). 2. Enrollment in clinical trials assessing new treatment strategies. 3. Bladder-preserving strategies in patients unsuitable or refusing RC.
Late BCG-relapsing: TaT1/HG recurrence > 6 months or CIS > 12 months of last BCG exposure	1. Radical cystectomy or repeat BCG course according to a patient's individual situation. 2. Bladder-preserving strategies 3. Enrolment in clinical trials assessing new treatment strategies.
LG recurrence after BCG for primary intermediate-risk tumour	1. Repeat BCG or intravesical chemotherapy. 2. Enrolment in clinical trials assessing new treatment strategies.

BCG = *bacillus Calmette-Guérin*; CIS = *carcinoma in situ*;
 LG = *low-grade*.

Follow-up

As a result of the risk of recurrence and progression, patients with NMIBC need to be followed-up. However, the frequency and duration of cystoscopy and imaging should reflect the individual patient's degree of risk.

Recommendations for follow-up of patients after transurethral resection of the bladder for NMIBC	Strength rating
Base follow-up of TaT1 tumours and carcinoma <i>in situ</i> (CIS) on regular cystoscopy.	Strong
Patients with low-risk Ta tumours should undergo cystoscopy at three months. If negative, subsequent cystoscopy is advised nine months later, and then yearly for five years.	Weak
Patients with high-risk and those with very high-risk tumours treated conservatively should undergo cystoscopy and urinary cytology at three months. If negative, subsequent cystoscopy and cytology should be repeated every three months for a period of two years, and every six months thereafter until five years, and then annually lifelong.	Weak
Perform cystoscopy at three months for patients with intermediate-risk Ta low-grade tumours. If negative, subsequent cystoscopy can be repeated every six months for two years, and then annually for ten years. The subgroup of intermediate-risk that are high grade should be followed up as high-risk.	Weak
Take regular and long-term upper tract imaging (computed tomography urography) for high-risk and very high-risk tumours.	Weak
Perform endoscopy under anaesthesia and bladder biopsies when office cystoscopy shows suspicious findings or if urinary cytology is positive.	Strong

During follow-up in patients with positive cytology and no visible tumour in the bladder, mapping biopsies or PDD-guided biopsies (if equipment is available) and investigation of extravesical locations (CT urography, prostatic urethra biopsy) are recommended.	Strong
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This short booklet text is based on the more comprehensive EAU Guidelines (978-94-92671-23-3) available to all members of the European Association of Urology at their website, <http://www.uroweb.org/guidelines/>.